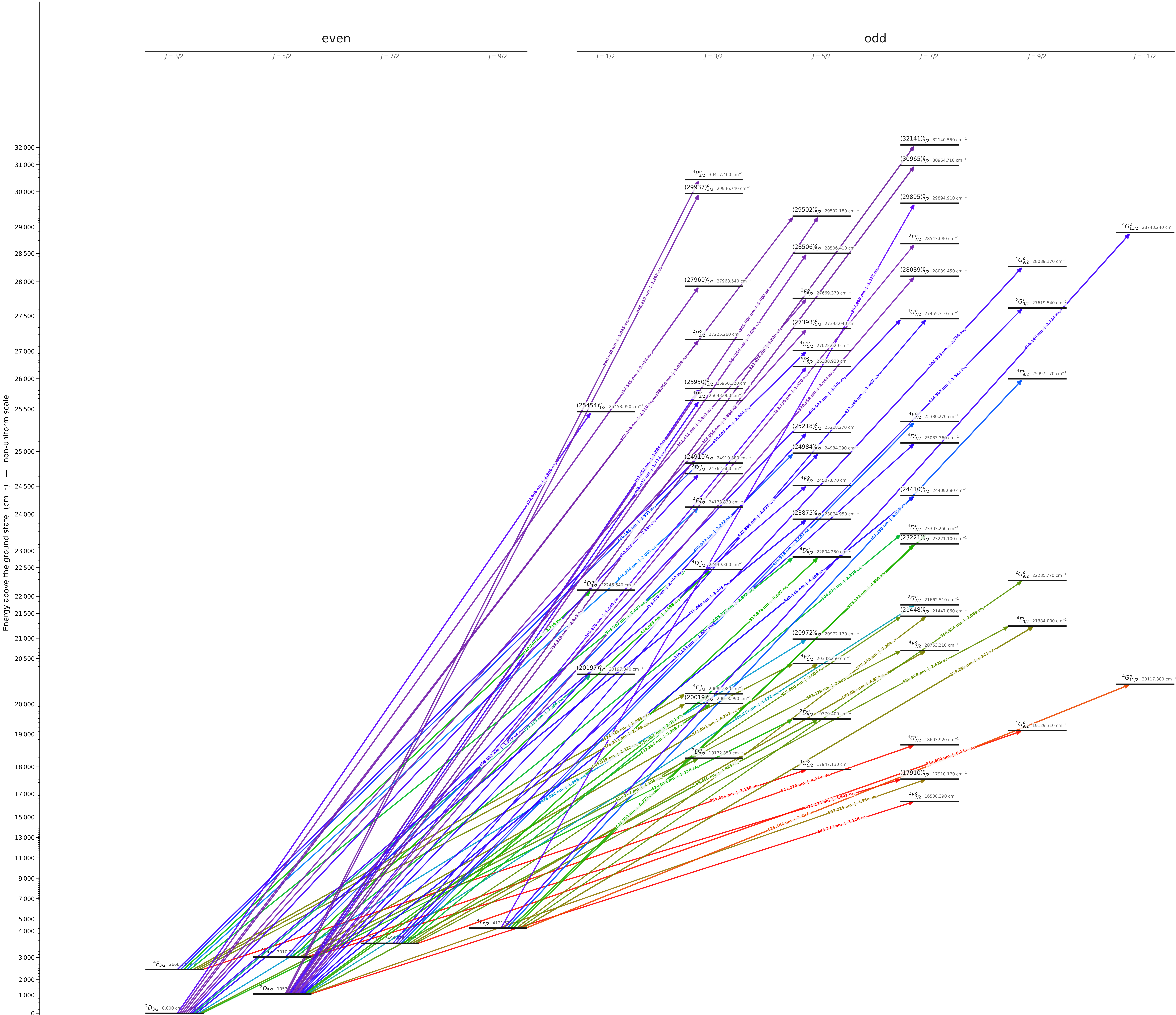
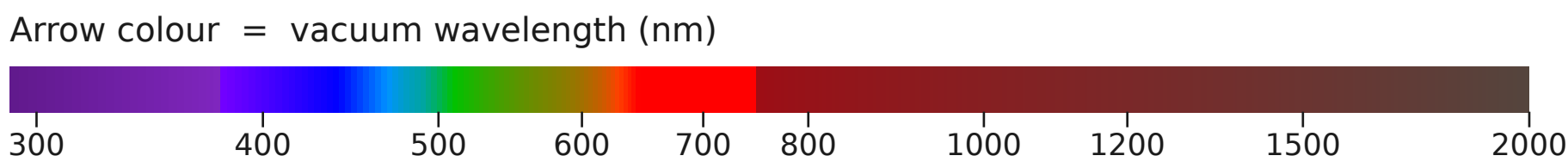


Lanthanum

La (La I). Strongest classified transitions below 2 μm (NIST): 62 levels, 70 lines.

La⁺ ionisation limit 44981.00 cm⁻¹ = 5.57693 eV (222.316 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U||er||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er J'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
² D _{5/2} - (30965) ⁰ _{1/2}	334.3191	334.2230	896.7256	2.823	5.400e+07	-	-	
² D _{3/2} - (27969) ⁰ _{3/2}	357.5446	357.4425	838.4757	2.928	9.500e+07	-	-	
² D _{5/2} - (28506) ⁰ _{5/2}	364.2557	364.1519	823.0276	3.609	9.100e+07	-	-	
² D _{3/2} - (25454) ⁰ _{1/2}	392.8663	392.7551	763.0902	2.359	9.300e+07	-	-	
² D _{5/2} - (25950) ⁰ _{5/2}	401.6523	401.5388	746.3980	2.884	6.500e+07	-	-	
⁴ F _{9/2} - ⁴ G _{11/2}	406.1463	406.0316	738.1390	4.714	5.600e+07	-	-	
⁴ F _{7/2} - ⁴ G _{9/2}	406.5926	406.4778	737.3289	3.786	4.320e+07	-	-	
² D _{3/2} - (23875) ⁰ _{3/2}	418.8490	418.7310	715.7530	3.463	5.510e+07	-	-	
² D _{5/2} - (24410) ⁰ _{5/2}	428.1460	428.0256	700.2107	4.199	5.690e+07	-	-	
⁴ F _{8/2} - ⁴ F _{8/2}	457.1304	457.0023	655.8139	4.523	4.340e+07	-	-	
² D _{3/2} - (20197) ⁰ _{1/2}	495.1147	494.9766	605.5010	3.284	9.000e+07	-	-	
⁴ F _{3/2} - ⁴ D _{1/2}	510.7656	510.6233	586.9472	3.716	1.050e+08	-	-	
⁴ F _{5/2} - ⁴ D _{3/2}	514.6850	514.5417	582.4775	4.698	8.200e+07	-	-	
⁴ F _{7/2} - ⁴ D _{5/2}	517.8738	517.7296	578.8910	5.807	8.200e+07	-	-	
⁴ F _{9/2} - ⁴ D _{7/2}	521.3306	521.1854	575.0525	5.273	4.970e+07	-	-	
² D _{3/2} - ² D _{5/2}	550.2866	550.1337	544.7934	4.104	5.120e+07	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).